

Ning Wei

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EDUCATION

University of Illinois at Urbana-Champaign | Grainger College of Engineering

Expected May 2027

Bachelor of Science, Computer Engineering (James Scholar) (Dean's list 2024/2025)

GPA : 3.85/4.0

Beijing No.4 high school

September 2020 - May 2023

TECHNICAL SKILLS

Programming Languages: C, C++, Python, Java, RISC-V Assembly, LC-3 Assembly

Development Tools: Git, Docker, FastAPI, React, Flutter, MySQL, PyTorch, MATLAB, Siemens NX, Vivado, Vitis

Hardware & Embedded Systems: SystemVerilog/Verilog, PlatformIO, Altium Designer, FPGA circuit design

Spoken Languages: Mandarin and English

RESEARCH EXPERIENCE

Biosensors Lab (Advisor: Prof. Viktor Gruev) | UIUC ECE Department

Champaign, IL, U.S.

Undergraduate Researcher

January 2026 - present

- AR-Assisted Tumor Imaging: Developing an AR-based intraoperative imaging pipeline using XR headsets to capture wavelength-specific responses and perform real-time image processing for enhanced visualization of tumor regions.
- Laser Illumination Control System: Developing an PCB-based embedded laser diode control system with constant-current driver and TEC thermal regulation, developed embedded firmware using an ESP32 controller to manage power domains, safety interlocks, and illumination control for biomedical imaging experiments.
- 3D-Printed Optical Tissue Phantoms: Fabricating tissue-mimicking optical phantoms using resin-based 3D printing with scattering and absorbing additives to replicate biological optical properties for SFDI (Spatial Frequency Domain Imaging) experiments.

Assured & Trustworthy AI Research Lab (Advisor: Prof. Huan Zhang) | UIUC CS Department

Champaign, IL, U.S.

Undergraduate Researcher

January 2026 - present

- LLM Coding Benchmark Pipeline: Developed an automated evaluation pipeline for large language models on coding benchmarks, integrating dataset management, model inference, sandboxed code execution, and execution-based correctness verification.
- AI4Science / LLMs for Analog Circuit Design: Investigating the capabilities of large language models for analog circuit design, including circuit netlist generation, SPICE Simulation, and iterative improvement through supervised fine-tuning and reinforcement learning.

WORK EXPERIENCE

Yuanli AI Co., Ltd. (Natural Language Processing Lab)

Beijing, China

Deep Learning Algorithm Intern

June 2025 - August 2025

- Fine-tuned on open-source large language base model for the education domain through continued pre-training, supervised fine-tuning, and sparse expert inference optimization, enhancing its performance in educational Q&A, problem-solving, and writing tasks.
- Developed a LLM based k12 math problem-solving visualization pipeline, transforming problems and solutions into structured plotting instructions, curating training data with reasoning steps and executable drawing code, and implementing a rendering engine to visualize the model's solution process.

SuperMap Software Co.,Ltd.

Beijing, China

AI Remote Sensing Data Engineering Intern

May 2025 - June 2025

- Researched and analyzed state-of-the-art semantic change detection models, including Siamese networks, early fusion, late fusion, and self-attention, to evaluate their advantages and applicability.
- Reproduced representative open-sources models (e.g., ChangeFormer, LeNet), built end-to-end training pipelines, and benchmarked performance across multiple datasets to support model optimization and deployment.

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IQIYI Inc.

Beijing, China

Machine Learning Engineering Intern

June 2024 - August 2024

- Researched and optimized existing Text2Video models, analyzing Diffusion Transformer and Pose-Controlled Generation for skeletal pose-guided video synthesis.
- Developed pre-processing scripts using computer vision and key-point detection to generate annotated video datasets and high-quality training data for pose recognition.
- Optimized the OpenSora-based training pipeline, integrating Control-Net for precise video generation using text and skeletal pose inputs, enhancing motion control and consistency.

University of Illinois at Urbana-Champaign Computer Science Department

Champaign, IL

CS124 Course Associate

January 2024 - June 2024

- Tutored CS124 Java students, assisted with assignments, and managed the online tutoring platform.

Timeloft Technology Company

Beijing, China

Software Engineer Intern

May 2023 - August 2023

- Optimized Business Middle Office architecture by modularizing business logic, implementing dependency injection, and integrating CI/CD pipelines, improving system stability, scalability, and development efficiency.

PROJECT HIGHLIGHTS

Time Series-Based Stock Market Prediction Model (Matlab, Python)

July 2024 - Present

- Modeled stock market trends using LSTM and Transformer architectures to capture nonlinear dynamics and long-term dependencies for improved forecasting accuracy.
- Built high-quality datasets with technical indicators and diverse training strategies to optimize prediction models and support risk-balanced trading and investment decisions.

FPGA-Based HDMI Graphics Gaming System (SystemVerilog, C, MicroBlaze)

August 2025 - December 2025

ECE 385 Digital Systems Laboratory Final Project

- Designed a real-time FPGA graphics system implementing a 640×480 HDMI pipeline with tile-based background rendering, sprite overlay, and obstacle scrolling entirely in SystemVerilog under strict video timing constraints.
- Implemented MicroBlaze-controlled game engine in C to manage world state, player movement, collision logic, USB keyboard input, interfacing with custom hardware graphics modules through AXI4-Lite memory-mapped registers.

Custom RISC-V Operating System Kernel (C, RISC-V Assembly)

August 2025 - December 2025

ECE 391 Computer Systems Engineering Course Project

- Implemented a UNIX-like kernel supporting process creation, ELF loading, system calls, and a user-level shell, with full kernel-user privilege separation on RISC-V.
- Designed virtual memory management with page tables, trap and page fault handling, and Copy-on-Write fork semantics, and integrated block device I/O (VirtIO) for persistent storage support.

Full-Stack Course & Team Matching Platform (Python, FastAPI, MySQL, React)

August 2025 - December 2025

CS411 Database System Final Project

- Designed and implemented a full-stack web platform enabling course discovery, team formation, and profile management, with a FastAPI-based REST backend handling authentication, matching logic, and transactional updates, backed by a normalized MySQL schema deployed on Google Cloud SQL.
- Implemented complex relational queries, indexing strategies, triggers, and stored procedures to support multi-condition filtering and availability-based matching, and integrated the backend with a modular React frontend for end-to-end cloud-based system operation.

Network Technology Department Member, UIUC CSSA

February 2024 - Present

- Managed IT infrastructure for the UIUC Chinese Students and Scholars Association, providing technical support.
- Developed and maintained the official website, online service platforms, and event management system.